

Junior Makers · Week 2

Two-Player Reaction Timer Game

Who has the fastest fingers? Build it, code it, design it!

Ages 8-11

Arduino + OLED

Game design

Did you know?

In Formula 1, the race starts when five red lights go out. Drivers react in about 0.2 seconds!



PLAYER 1



PLAYER 2





Quick challenge!

Ready... steady... GO!



The ruler drop test

1. Your partner holds a ruler above your open fingers.
2. Without warning, they let go.
3. Catch it as fast as you can!
4. The lower the number you catch it at, the **faster** your reaction.



Racing drivers react to start lights



Goalkeepers dive for penalty kicks



Gamers tap faster than opponents



Can we build a machine that measures reactions fairly, and turn it into a game?





Learning aims

Today I will learn to...



1  Wire **2 buttons**, 2 LEDs, a buzzer and an **OLED** screen

2  Use **random numbers** and **timing** in milliseconds

3  Use **variables** to keep score

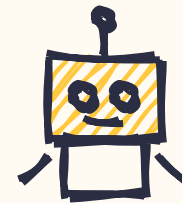


4  Design rules like a **game developer**

5  **Playtest** and improve our game

Keywords

reaction time · millisecond · random ·
variable · game loop · game state ·
playtest



Watch and think

Video time!



The Reaction Time Test!

Operation Ouch (BBC) • about 6 minutes • youtu.be/kj-L-6MfaE0

Watch for...

- 1 What is **reaction time**?
- 2 Which part of your body sends the 'press now' message?
- 3 What can make your reactions **faster** or slower?

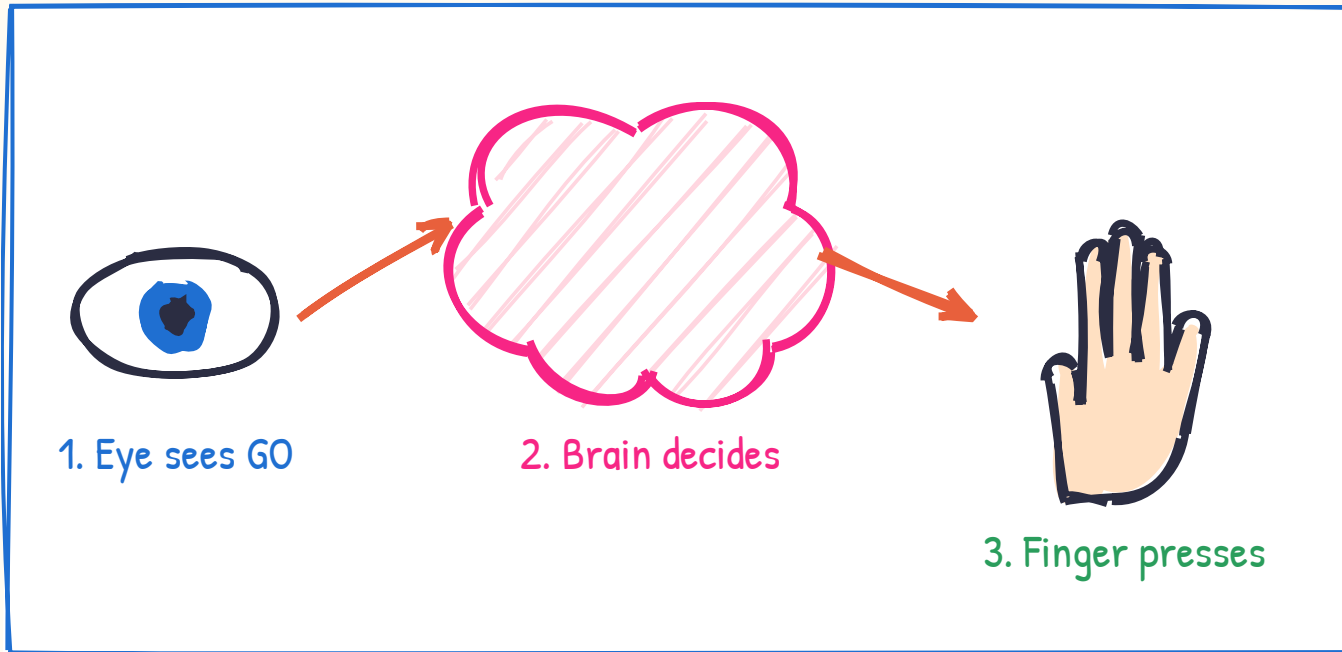


After the video: Do the ruler-drop test with your partner. Who caught it at the lowest number?



The science

What is reaction time?



Measured in milliseconds

1 second = **1000 ms**

Most people react to a light in about **250 ms** (a quarter of a second).

Practice and sleep can make you faster!



In real life

In athletics, a start faster than 0.1 seconds counts as a false start: nobody can react that fast, so they must have guessed!





Put on your game developer hat

Every great game has...



Goal

What players try to do



Rules

What is allowed, what is not



Challenge

Hard enough to be fun



Feedback

Lights, sounds, scores



Win condition

How the game ends



Fair play

Same chance for everyone



In real life

Game studios write all of this in a Game Design Document before anyone writes code.



Think of your favourite game. Can you name its goal, rules and win condition?





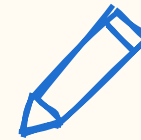
Design first

Our Game Design Document (GDD)



Your turn
Invent a better name for your version of the game!

Game name	Fastest Fingers
Players	2 (one button each)
Goal	Press your button first after the green GO light
Rules	Press before GO = FALSE START, the other player gets the point
Feedback	Buzzer beep, LEDs, time and score on the OLED
Win	First to 3 points (best of 5 rounds)





How games think

Game states: what is happening right now?



1. WAITING

Show scores. Press any button to start



2. GET READY

Random wait: 2 to 5 seconds



3. GO!

Green LED on, timer starts



4. WINNER

Show time, add a point, beep



pressed too early?

FALSE START!

Pressed during GET READY: red LED, other player scores, back to 1



A **state** is what the game is doing at this moment. The code checks the state and decides what to do next. After WINNER, it goes back to WAITING.

In real life

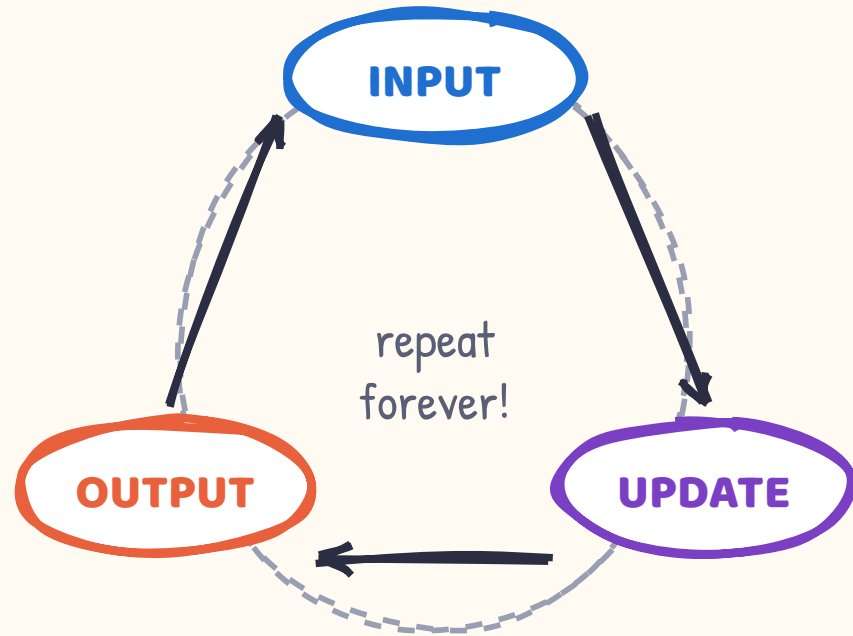
Video games switch between states too: menu, playing, paused, game over.





The heartbeat of every game

The game loop



INPUT

Read the buttons: did a player press?

UPDATE

Change the state, measure the time, add points

OUTPUT

Light the LEDs, beep, draw the scoreboard

In real life

Games like Minecraft run this loop about 60 times every second!



Our game console

Hardware we need (per pair)



Arduino Uno

The game console



2 buttons

The controllers



Green LED

GO signal



Red LED

False start



Buzzer

Sound effects



OLED screen

The scoreboard

Plus: breadboard, 2 × 220 Ω resistors (LEDs), 2 × 10 k Ω resistors (buttons), jumper wires, laptop.

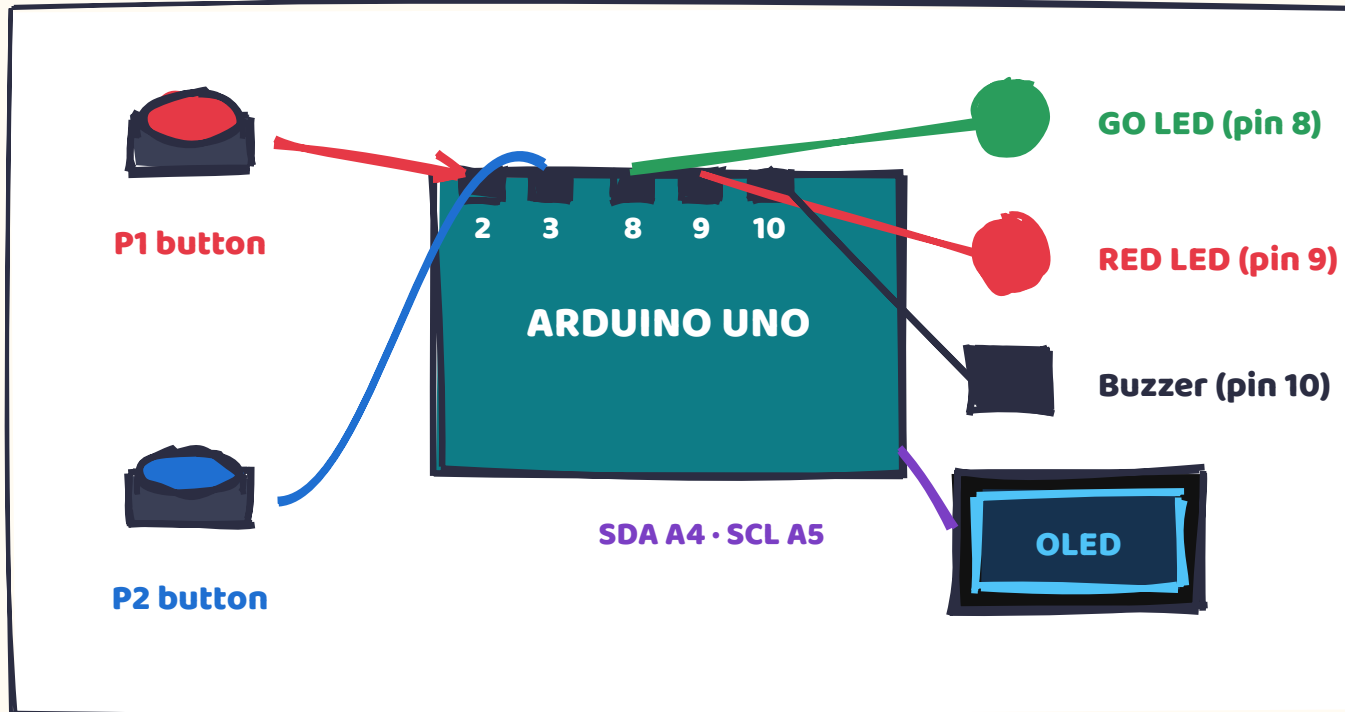
In real life

Game consoles are also computers with inputs (controllers) and outputs (screen, sound).



Build it

Wiring the game



P1 button	pin 2 (+ 10 kΩ to GND)
P2 button	pin 3 (+ 10 kΩ to GND)
Green GO LED	pin 8 (+ 220 Ω)
Red LED	pin 9 (+ 220 Ω)
Buzzer	pin 10
OLED	SDA → A4, SCL → A5, VCC → 5V, GND

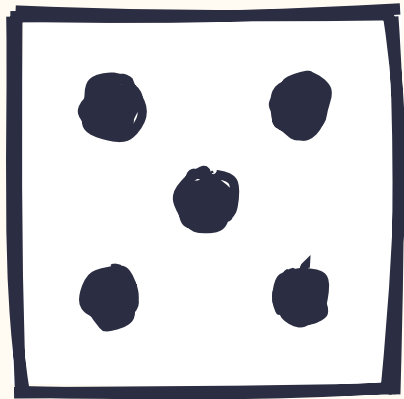
Test first

Build it in Tinkercad Circuits or Wokwi before using real parts.



No cheating!

Why the wait must be random



If GO always came after 3 seconds...

Players would count "1, 2, 3" and press without looking. That tests counting, not reactions!

So we let the computer pick

`random(2000, 5000)` picks a wait between 2 and 5 seconds, different every round.

Game design idea

Randomness makes games exciting and fair: dice, card shuffles, surprise enemies.

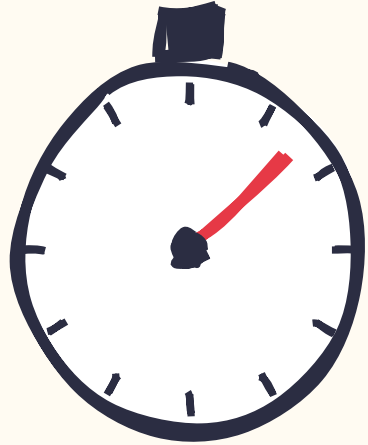
In real life

Online games use random numbers for loot boxes and where players spawn.



A built-in stopwatch

Measuring time with millis()



How it works

millis() tells us how many milliseconds since the Arduino switched on.

At GO: **startTime = millis()**

When pressed: **reaction = millis() - startTime**

Think

Why not use a wait block to time the player? (Hint: a wait freezes the whole program!)

Worked example

GO at 12,400 ms · press at 12,643 ms → **243 ms**





Code it

The game in ArduinoBlocks

**when Arduino starts**

set score1 to 0

set score2 to 0

forever

show 'Get ready...' on OLED

wait random 2000 to 5000 ms

if a button is already pressed → FALSE START

turn ON green LED, beep

set startTime to millis

wait until P1 or P2 pressed

set reaction to millis - startTime

add 1 to winner's score

show winner, time and scores

Colour key

Orange: variables (scores, time)**Yellow:** control (loops, waits)**Purple:** screen output**Teal:** lights and sound**Red:** the false-start check

Look inside

ArduinoBlocks can show the C++ code it writes.
Take a peek!





Show the score

Designing the OLED scoreboard

**FASTEST FINGERS****P1: 2 P2: 1****Winner: P1****Time: 243 ms****Best: 198 ms**

Variables = the game's memory

Variable	Stores
score1	Player 1 points
score2	Player 2 points
reaction	Time this round (ms)
bestTime	Fastest time so far



Game design tip

Players love seeing their best score. It makes them want to play again!





Work like a game studio

Playtest, get feedback, improve



Designer

Writes the rules



Programmer

Builds the code



Tester

Finds bugs, checks fairness



Sound artist

Designs beeps and tunes



Playtest checklist

- ☒ Is it fun? Would you play again?
- ☒ Is it fair to both players?
- ☒ Is the scoreboard easy to read?
- ☒ What is ONE thing we should change?



The golden rule

Change one thing, playtest again, keep what makes it better.





Level up!

Challenges for game designers



Level 1: Match mode

Best of 5 with a winner celebration tune



Level 2: Trick light

Sometimes the RED LED flashes: do NOT press!

Boss level

Make it 3 or 4 players with more buttons!



Level 3: Practice mode

One player tries to beat their best time



Level 4: Sound design

Different beeps for GO, win and false start



Show what you know

Recap and exit ticket

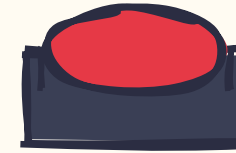
Exit ticket

- ☒ Why is the waiting time random?
- ☒ What does `millis() - startTime` tell us?
- ☒ Name the 4 game states in our game.
- ☒ What 3 steps does every game loop repeat?

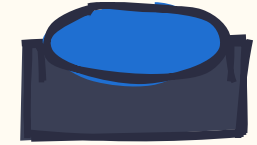


Class tournament!

Winners of each pair play the next round. Who is the class reaction champion?



P1



P2



Next week: a Mini Weather Station with an OLED display!